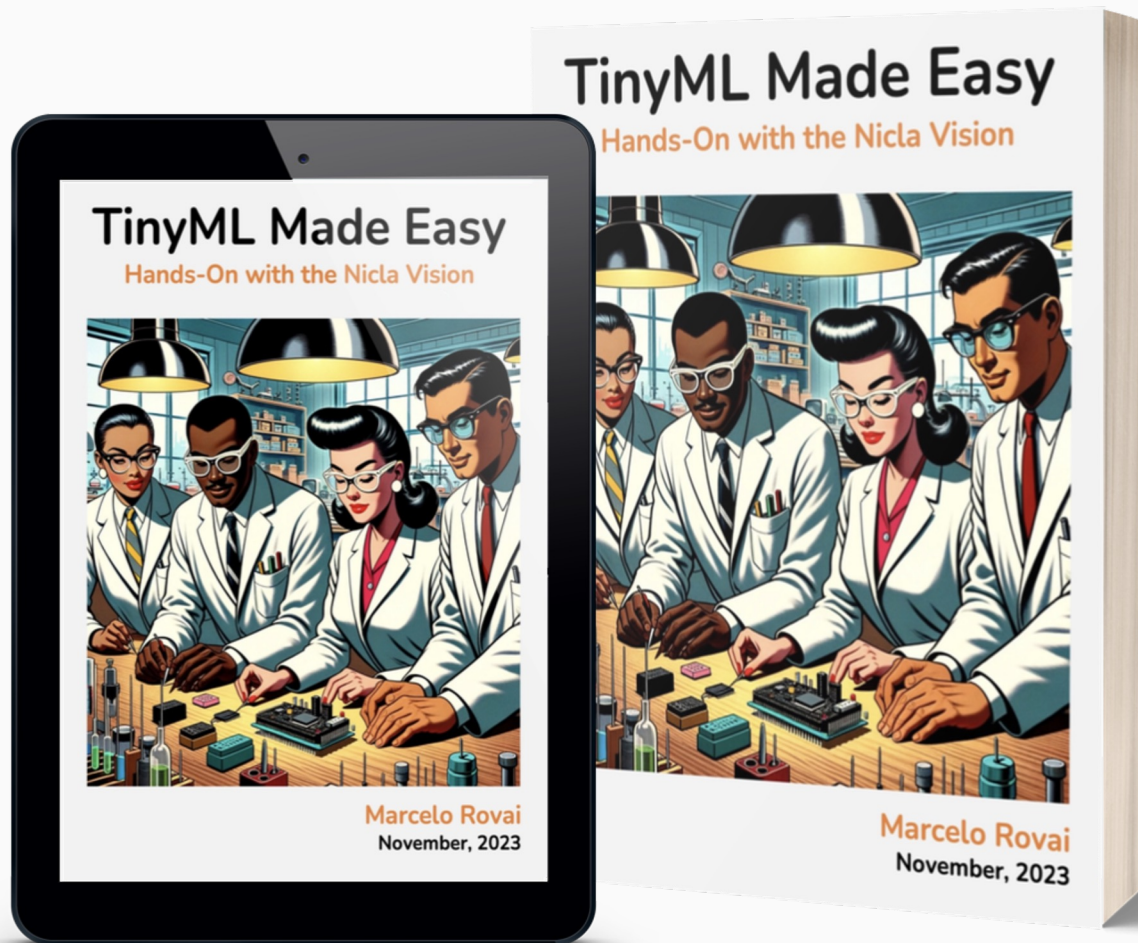


Getting to know the Arduino Nicla Vision

Marco Zennaro, PhD

The Abdus Salam International Centre for Theoretical Physics –Italy

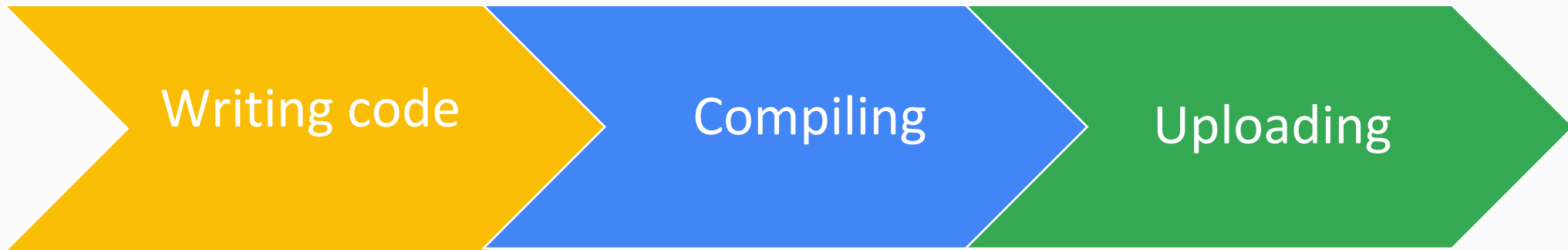


https://mjrovai.github.io/TinyML_Made_Easy_NiclaV_eBook/



What will we learn?

The workflow of writing and executing code on the Nicla Vision

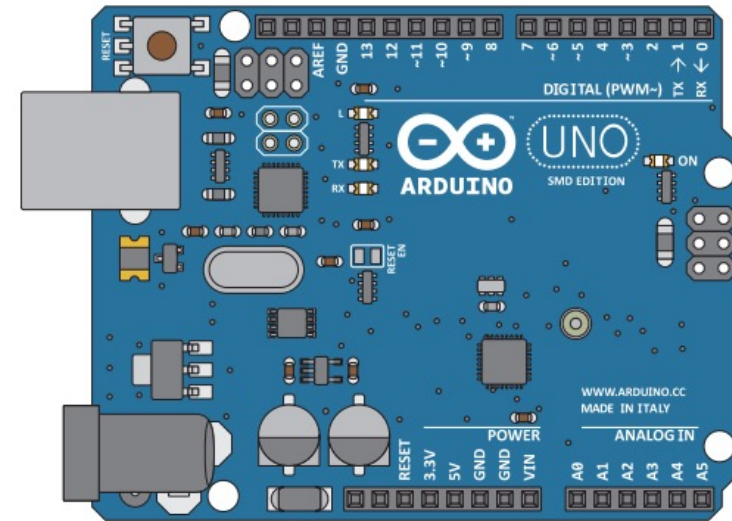


A computer for the physical world.

Able to read INPUTS – light on a sensor, a finger on a button, or a Twitter message.

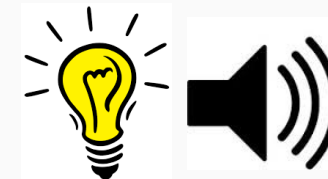
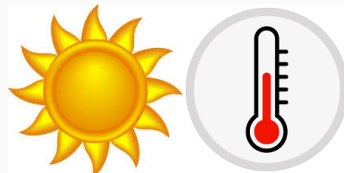
Turn it into OUTPUTS – activating a motor, turning on an LED, publishing something online.

Tell your Arduino what to do using the Arduino programming language.

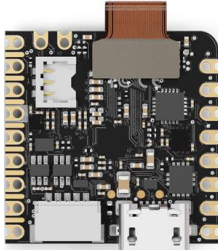
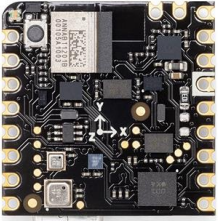
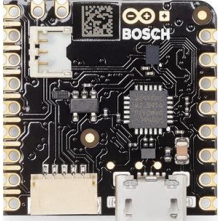
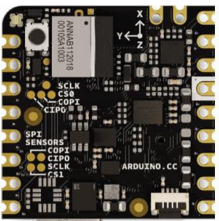


www.arduino.cc

Uses information from INPUTS to control various OUTPUTS.

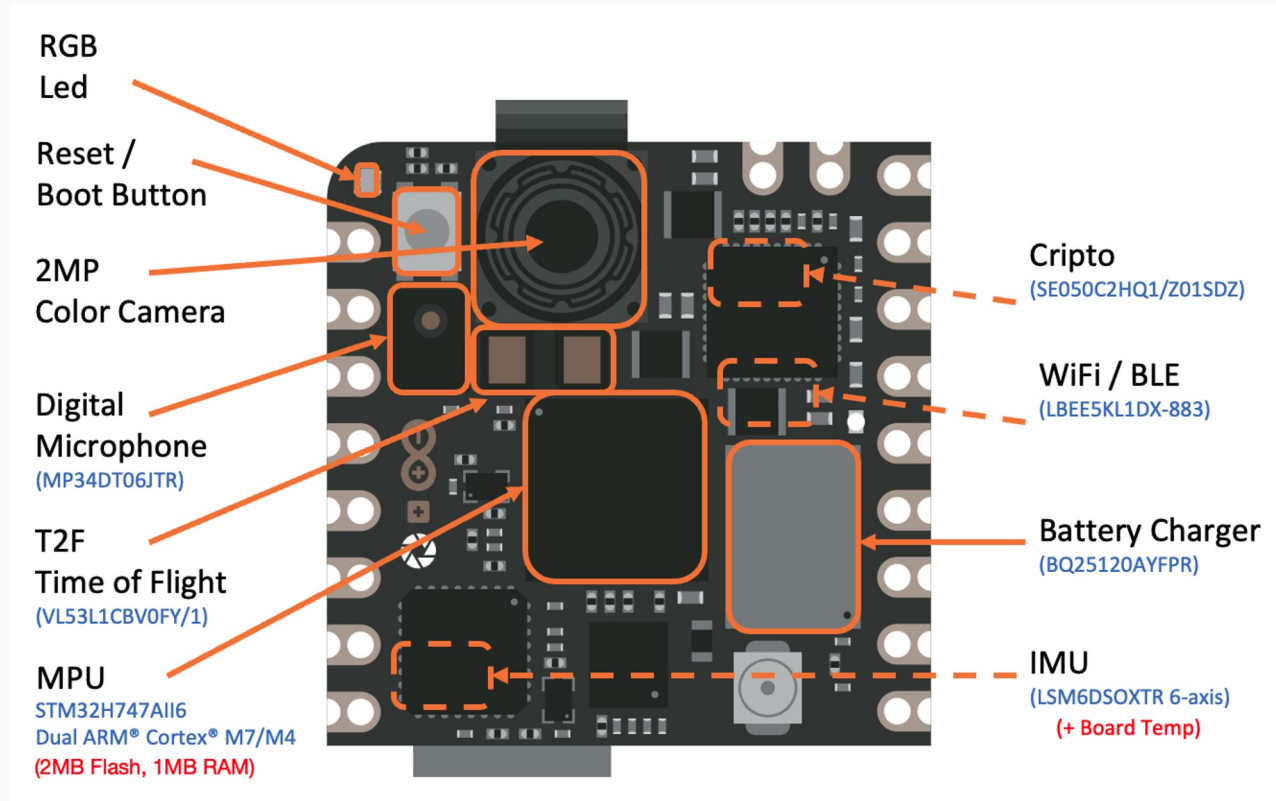


Arduino PRO Nicla Family

Board	Nicla Vision		Nicla ME		Nicla Voice	
	 (Top View)	 (Bottom View)	 (Top View)	 (Bottom View)	 (Top View)	 (Bottom View)
MPU	None		None		NDP-120	
MCU	ST STM32H747AI16		Nordic nRF52832		Nordic nRF52832	
Core1	Arm Cortex-M7 @480MHz		Arm Cortex-M4 @64MHz		Arm Cortex-M4 @ 64MHz	
Core2	Arm Cortex-M4 @240MHz		None		None	
Flash	2MB + QSPI 16MB		512KB + QSPI 2MB		512KB + QSPI 16MB	
SRAM	1MB		64KB		64KB	
PDM	MP34DT05		None		IM69D130	
IMU	LSM6DSOX		BHI260AP		BMI270	
Mag.	None		BMM150		BMM150	
Pressure	None		BMP390		None	
T & H	None		BME688		None	
ToF	VL53L1CBV0FY/1		None		None	
Camera	GC2145 (2MP)		None		None	
Security	NXP SE050C2		None		None	
BLE	BLE 4.2 (1DX-LBEE5KL1DX-883)		BLE 4.2		BLE 5.0 (ANNA-B1112)	
Connect	WiFi / USB		WiFi / USB		USB	
Charger	MAX1726REWL		BQ25120AYFPR		BQ25120AYFPR	
Size	22.86 mm x 22.86 mm		22.86 mm x 22.86 mm		22.86 mm x 22.86 mm	

MPU	Syntiant® NDP120 Neural Decision Processor™ (NDP): 1x Syntiant Core 2™ ultra-low-power deep neural network inference engine; 1x HiFi 3 Audio DSP; 1x Arm® Cortex® M0 core up to 48 MHz, 48K SRAM.
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Arduino Nicla Vision

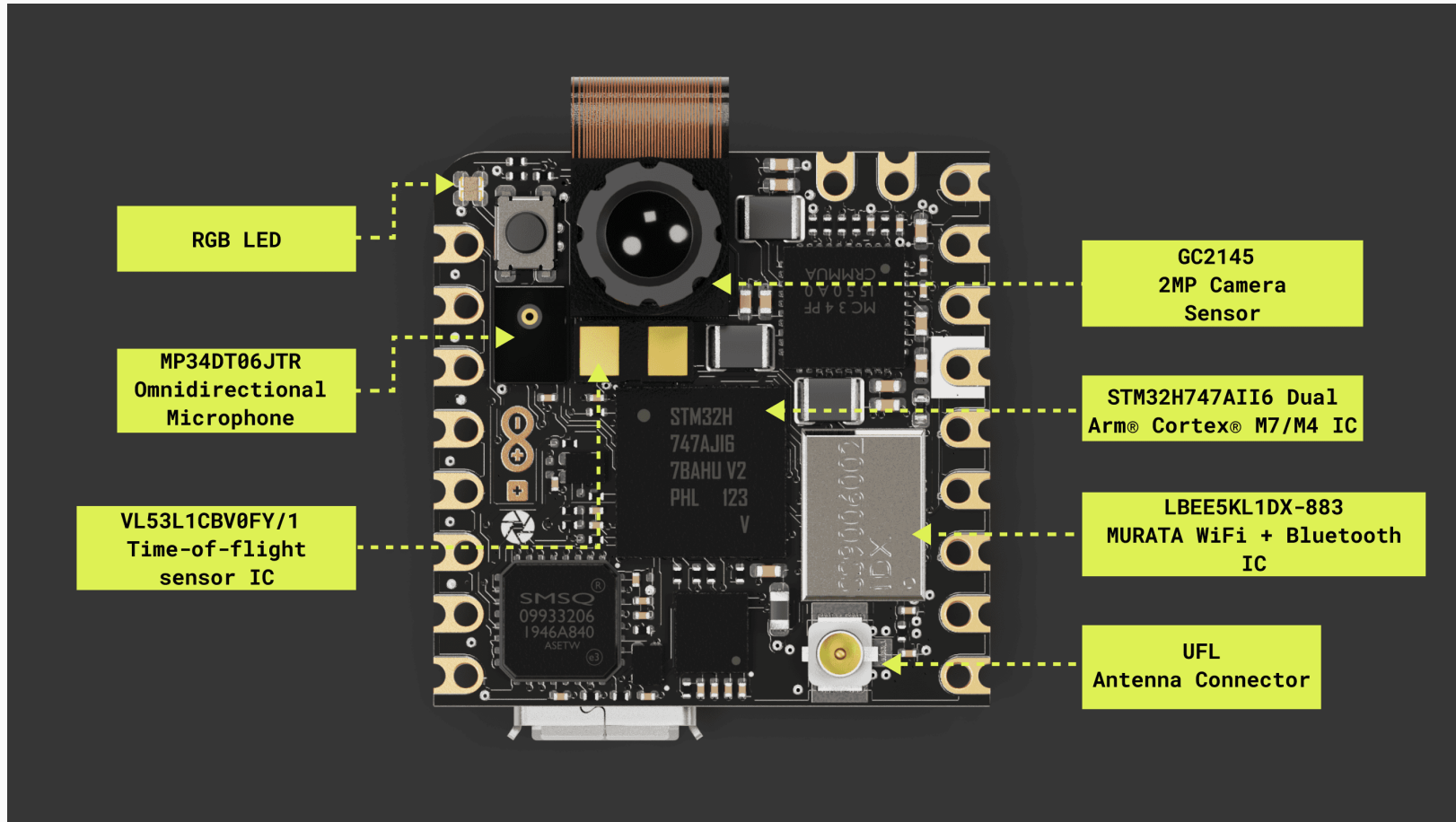


Arduino Nicla Vision

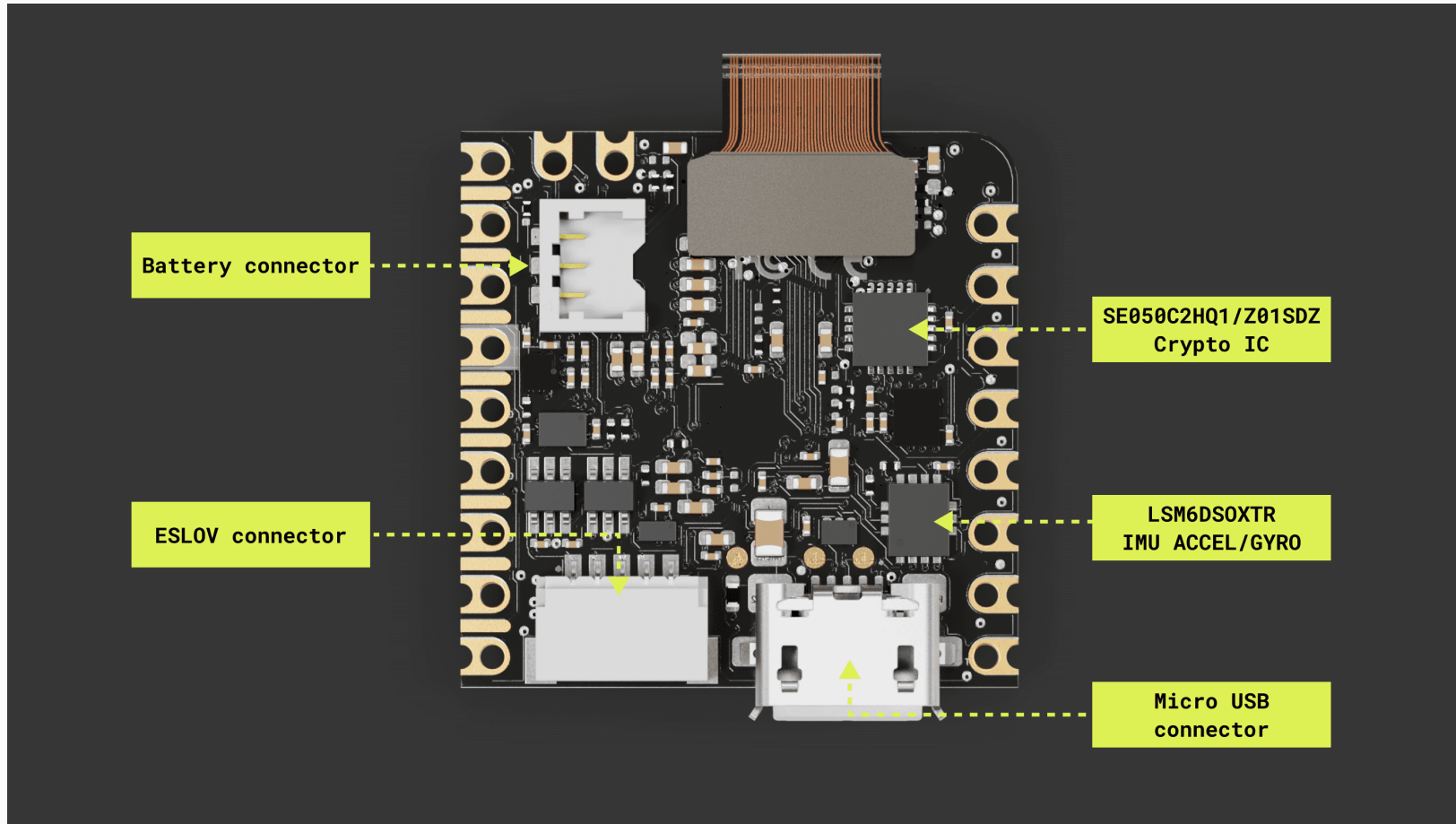
The central processor is the dual-core STM32H747, including a Cortex® M7 at 480 MHz and a Cortex® M4 at 240 MHz.

The STM32H747 is only 1MB, shared by both processors. This MCU also has incorporated 2MB of FLASH, mainly for code storage.

Arduino Nicla Vision



Arduino Nicla Vision



Edge
Device



& Sensors



Arduino programming

The programming language is based on Wiring and in terms of syntax (almost) identical to C++.

This is the template of a basic Arduino program:

```
void setup()
{
  Initialize variables, open USB, open WiFi, etc
}
```

SETUP
(once)

```
void loop()
{
  Perform some action

  Wait for a certain number of msecs or wait for an alarm
}
```

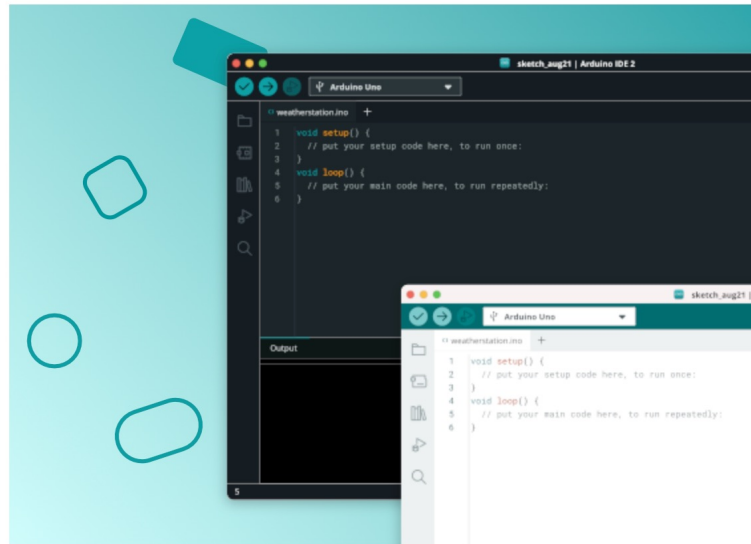
LOOP
(forever)

Arduino IDE

IDE= Integrated development environment

Arduino IDE is Open Source

Go to: <https://www.arduino.cc/en/software>



Arduino IDE 2.3.8

[Release notes](#)

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger. For more details, check the [Arduino IDE 2.0 documentation](#).

macOS Intel 10.15 Catalina or newer (64-bit)

[DOWNLOAD](#)



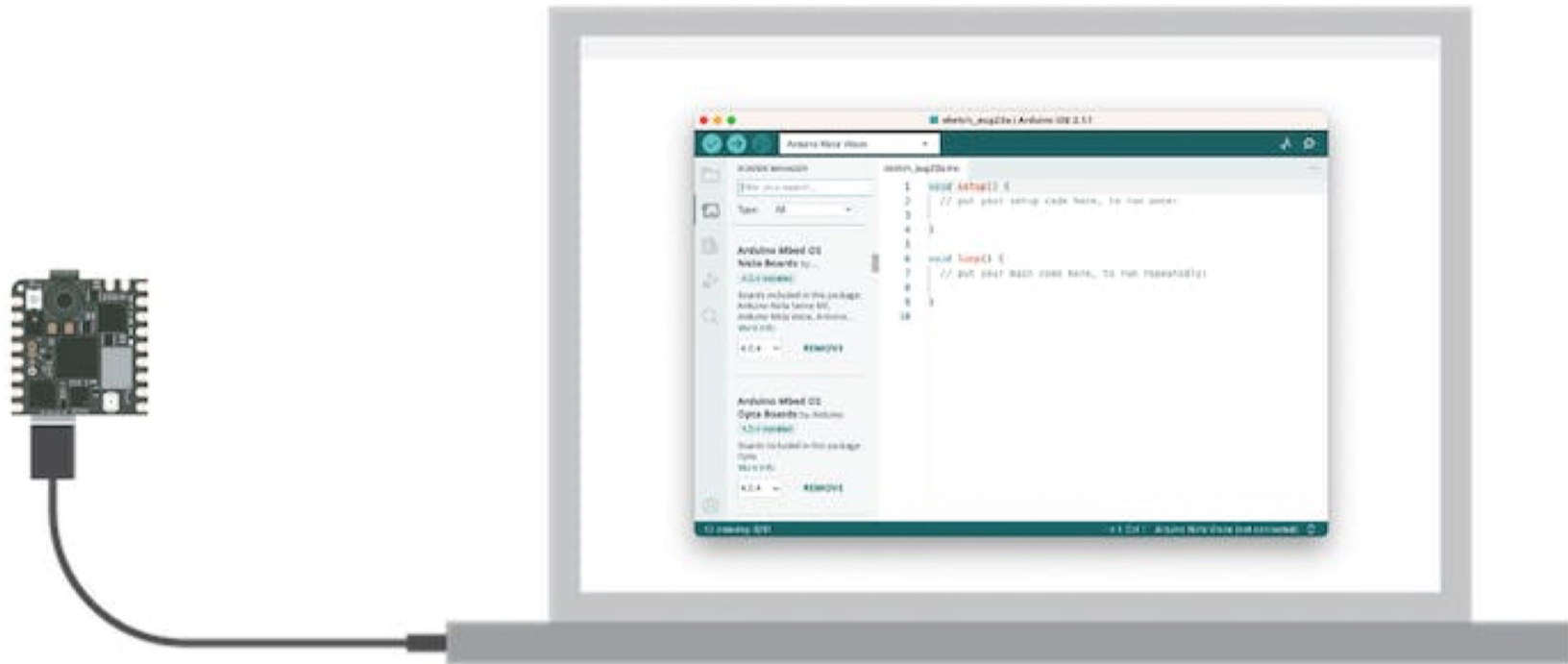
Nightly Builds

Download a preview of the incoming release with the most updated features and bugfixes.

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

Arduino IDE

Start connecting the board (*microUSB*) to your computer:



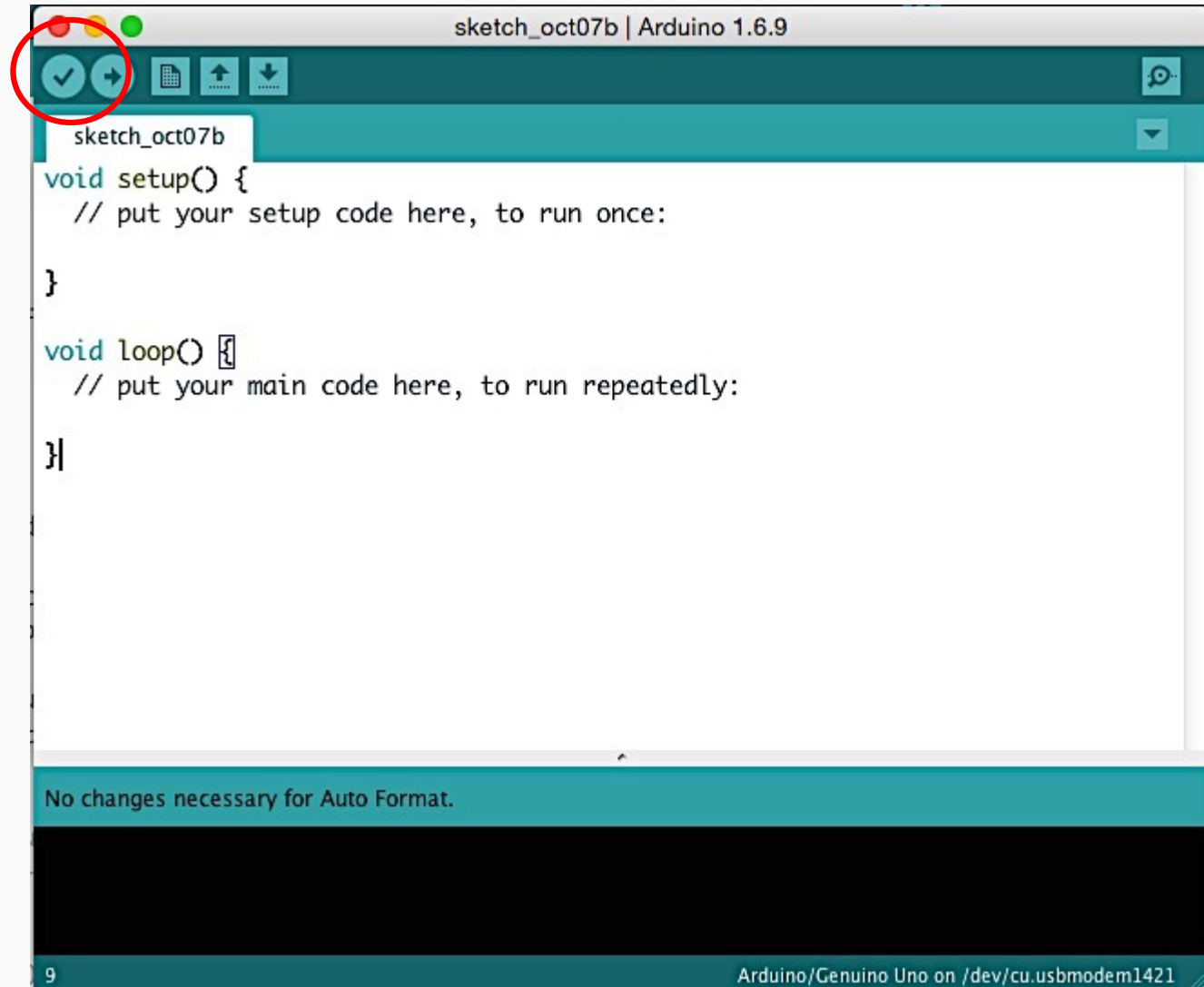
Installing Arduino libraries

Install the Mbed OS core for Nicla boards in the Arduino IDE. Having the IDE open, navigate to Tools > Board > Board Manager, look for Arduino Nicla Vision on the search window, and install the board.

Next, go to Tools > Board > Arduino Mbed OS Nicla Boards and select Arduino Nicla Vision. Having your board connected to the USB, you should see the Nicla on Port and select it.

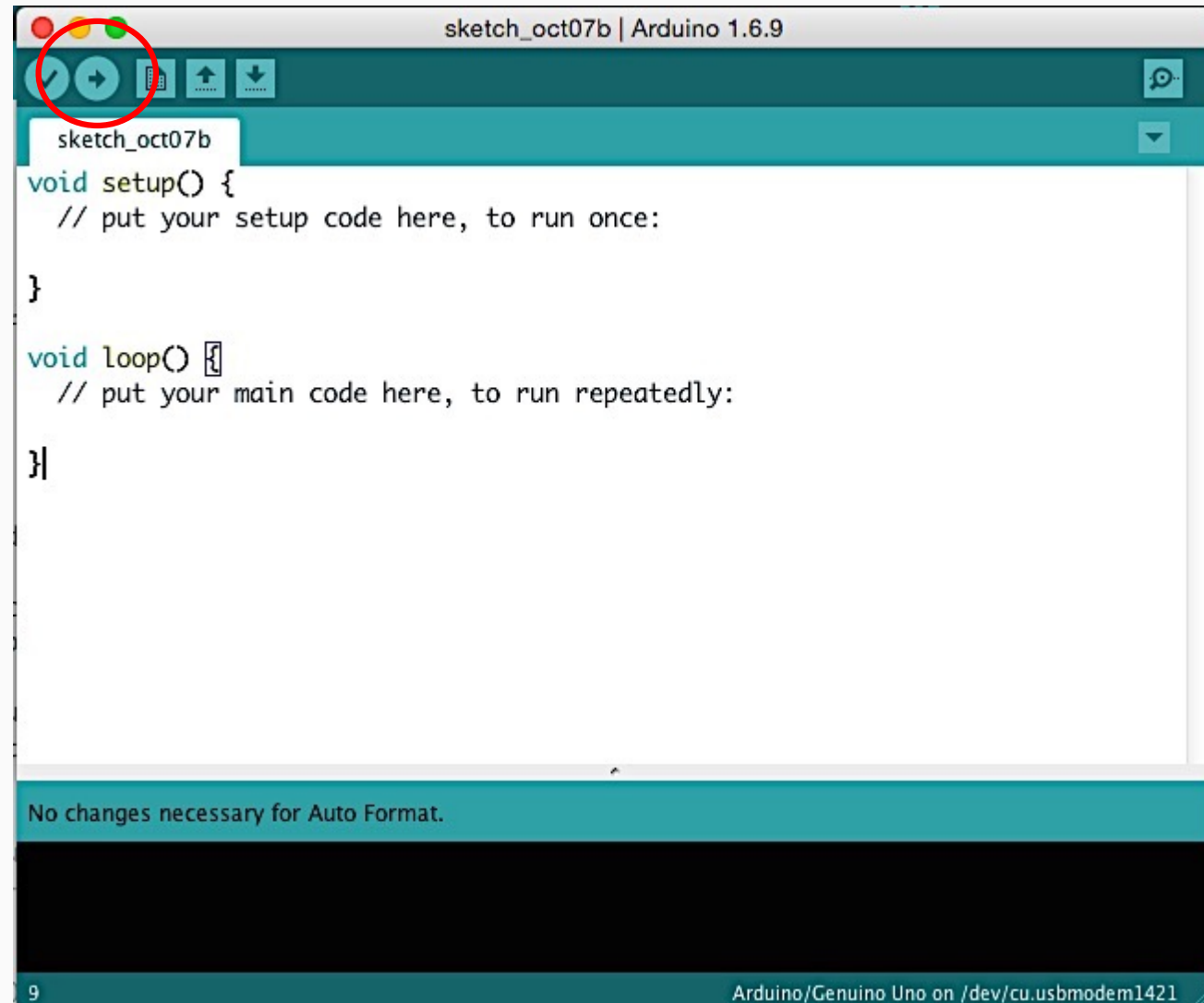
Verify:

Compiles and approves your code. Will catch errors in syntax.



Upload:

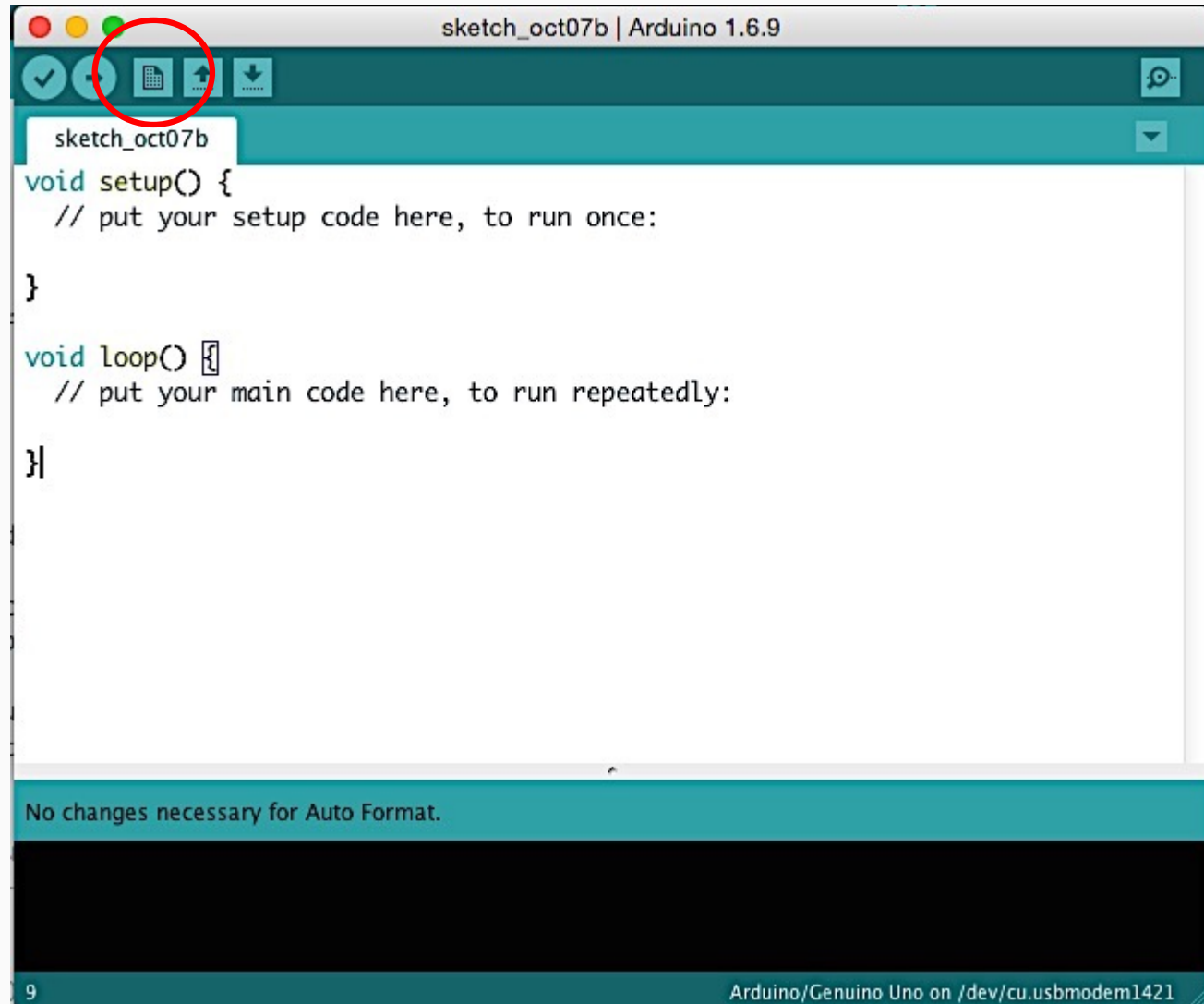
Sends your code to the Arduino Board. When you click it, you should see lights on your board blink rapidly.



KNOW the Arduino GUI

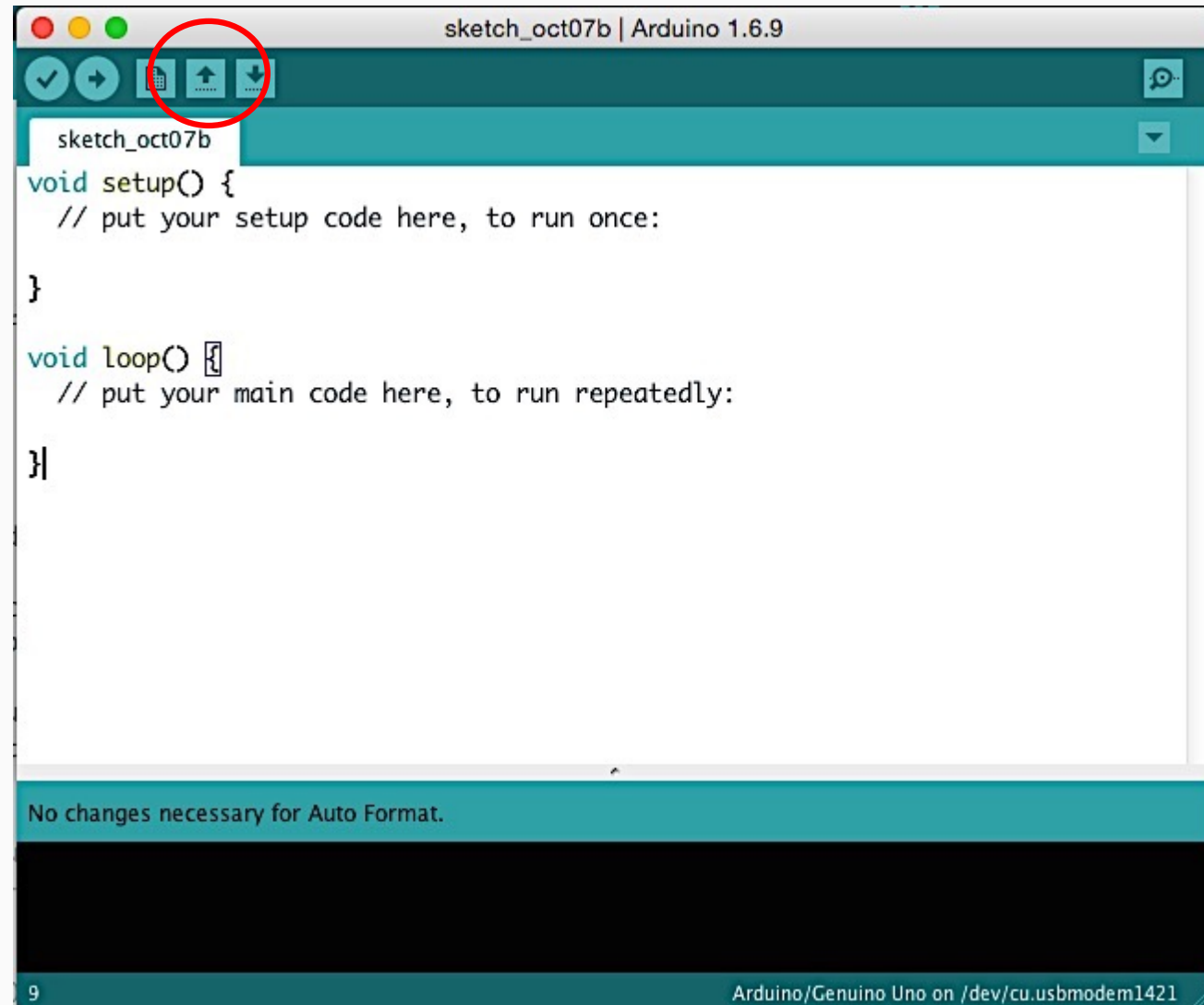
New:

Opens up a new code window tab.

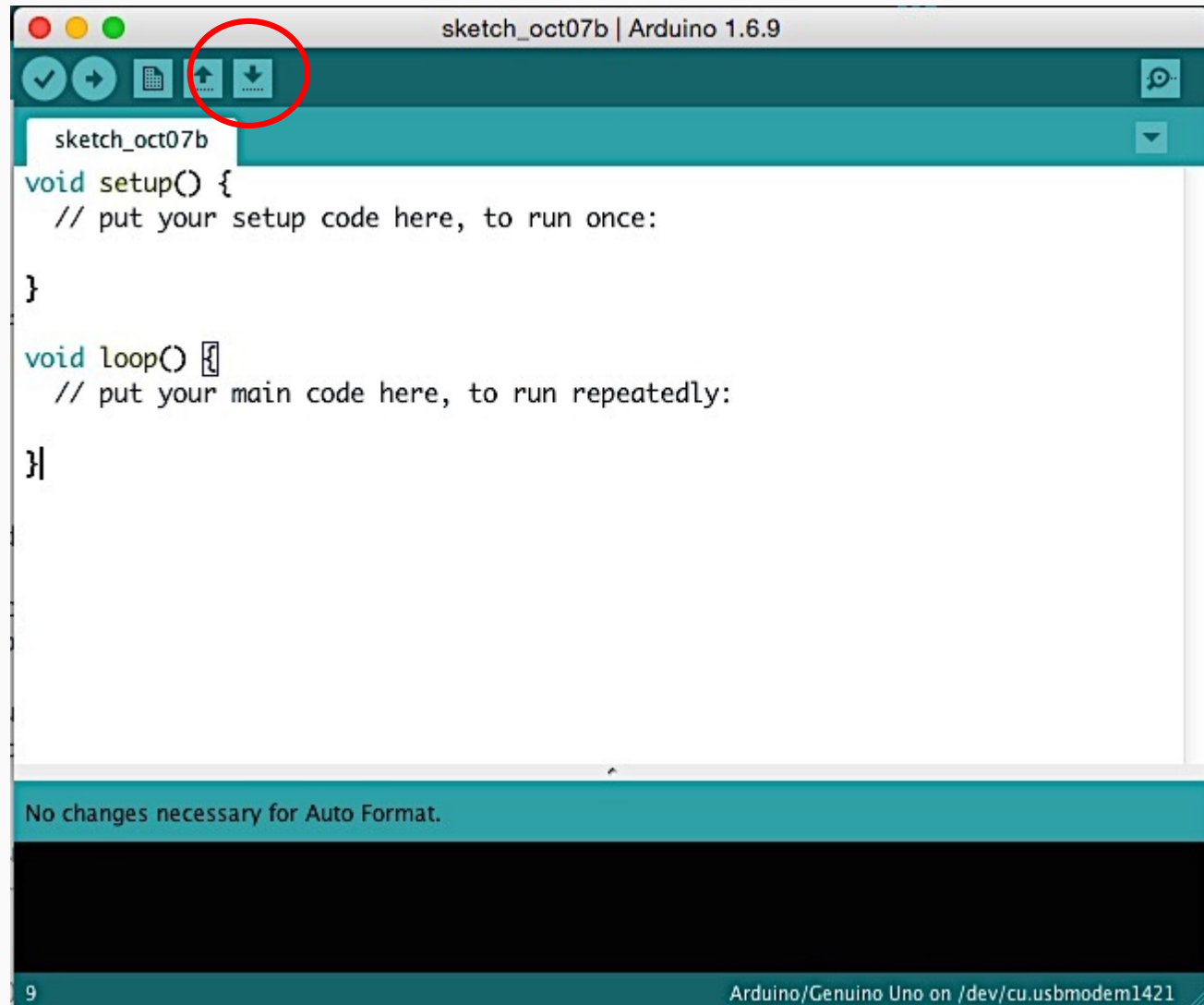


Open:

Opens an existing sketch, which is where you write your code.



Save:
Saves the
currently open
sketch.



Serial

Monitor:

Opens a window that displays any serial info the Arduino Board is transmitting. Very useful for debugging.



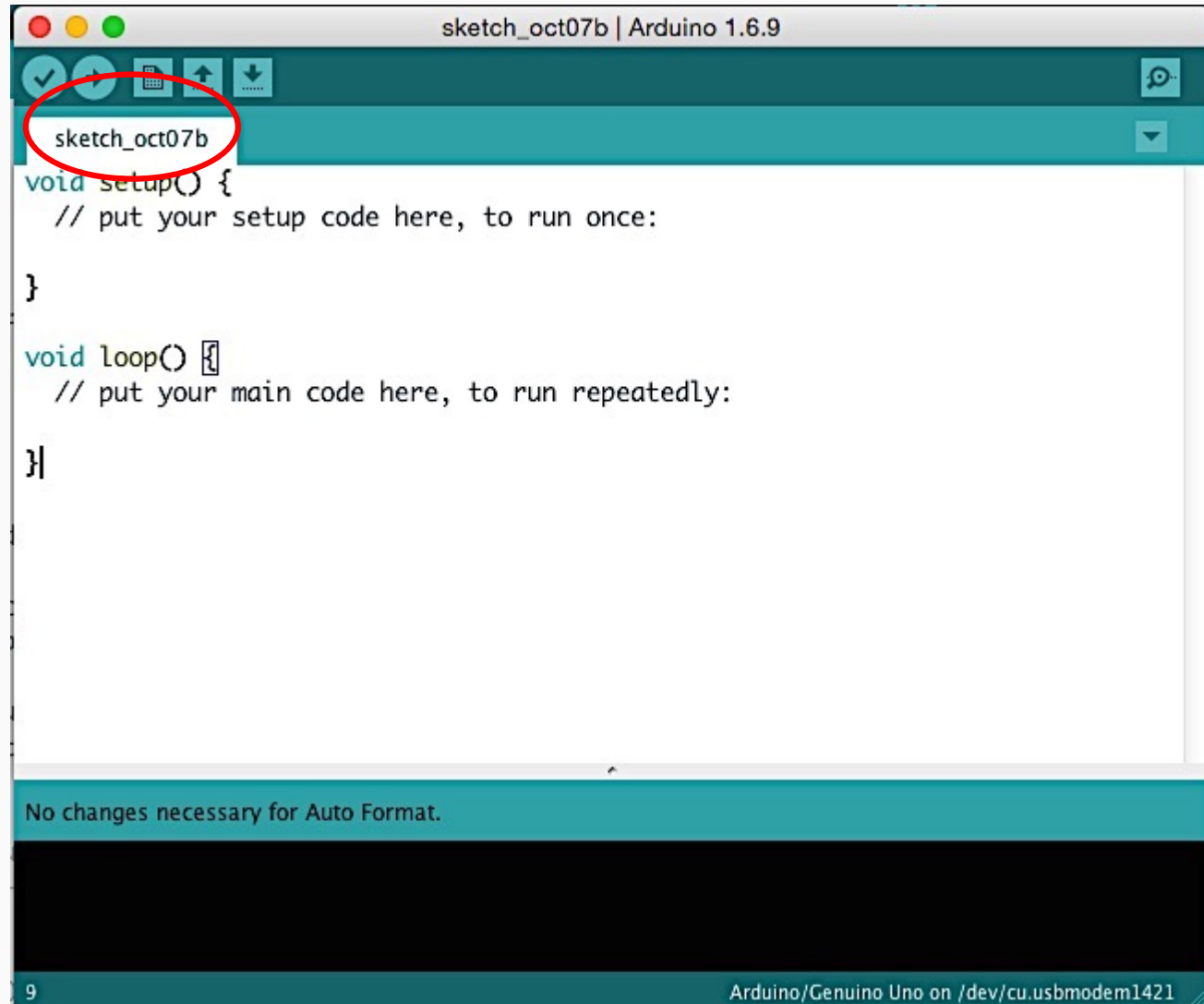
**Serial
Plotter:** Opens
a window that
displays in
graphical form
any serial info
the Arduino
Board is
transmitting



Sketch

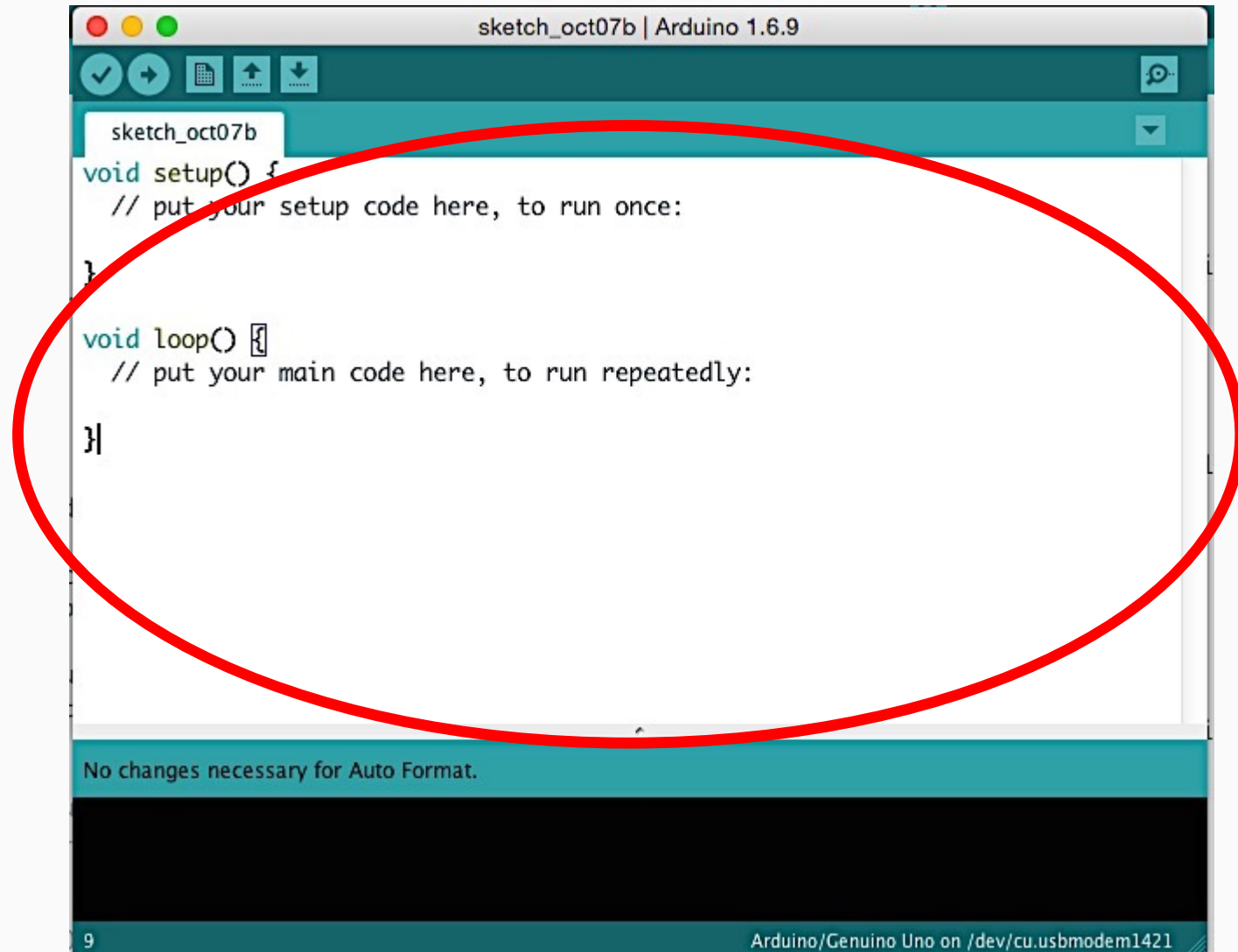
Name:

Name of the sketch you are currently working on.



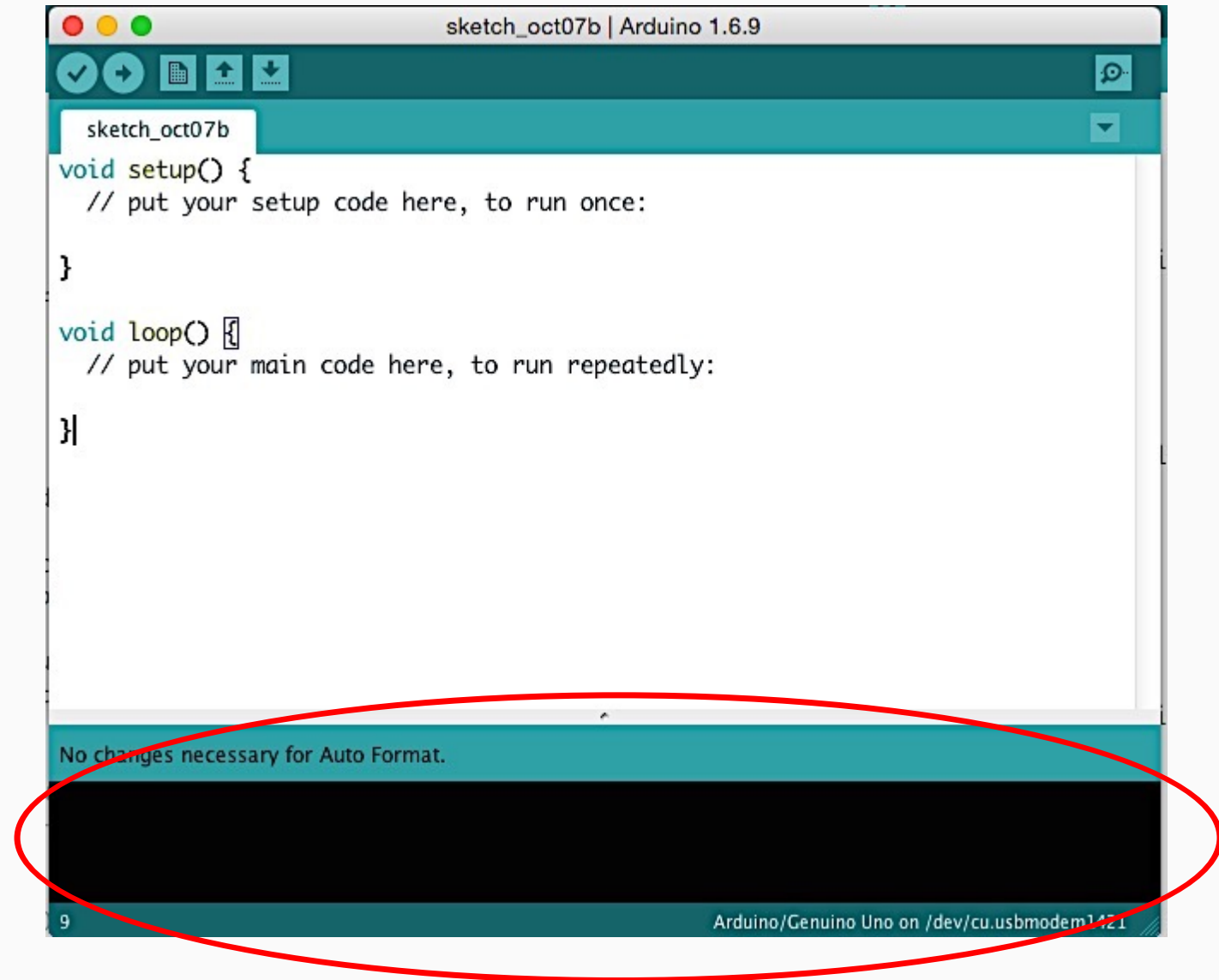
Code Area:

Area where you compose the code for your sketch.



Message Area:

Where the IDE tells you if there were any errors in your code.



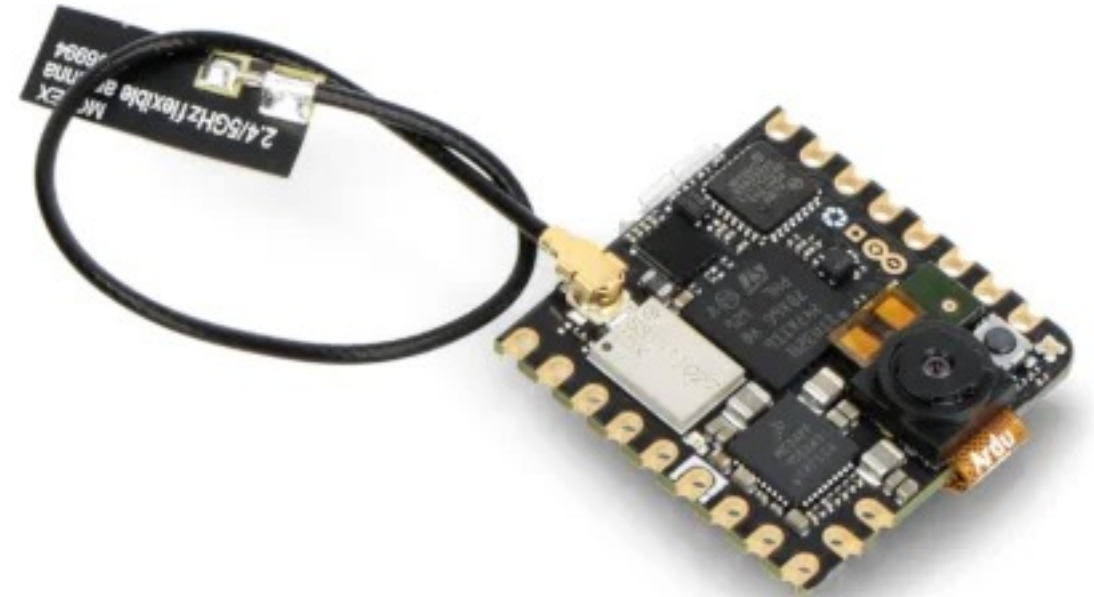
WiFi/BLE antenna

The Nicla Vision ships with an external 2.4/5GHz antenna for BLE and WiFi connectivity.

Install this antenna by carefully aligning the connector and applying sufficient pressure for it to snap into place.

Note that once connected, the antenna should only be removed using a specialised tool, in order not to damage the connector.

Using WiFi without the antenna might damage the board!



Warm-up Task



Hello world for Arduino

Open the Blink sketch on Examples/Basic and run it using the IDE Upload button. You should see the Built-in LED (green RGB) blinking, which means the Nicla board is correctly installed and functional!

Exercise:

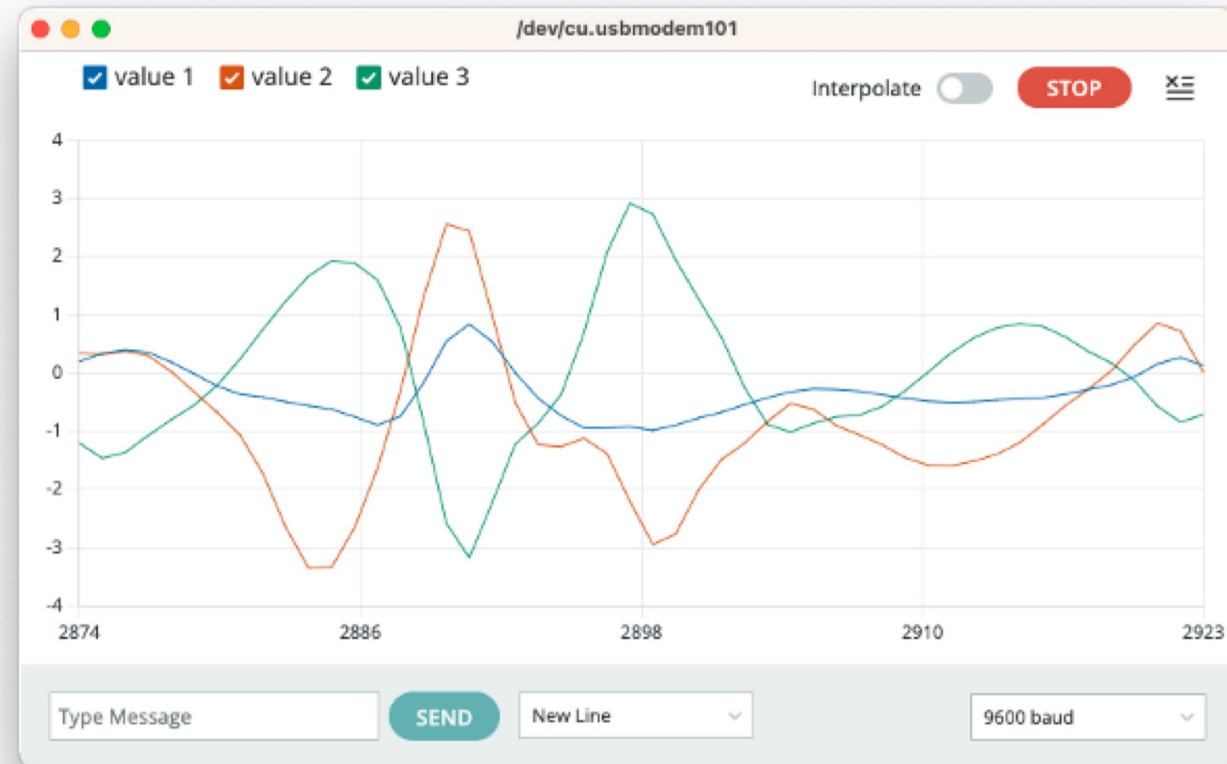
Make the LED blink for a different amount of time

Make the LED blink as fast as possible!

Accelerometer test

Before testing the IMU, it will be necessary to install the LSM6DSOX library. For that, go to Library Manager and look for LSM6DSOX. Install the library provided by Arduino:

Next, go to Examples > Arduino_LSM6DSOX > SimpleAccelerometer and run the accelerometer test (you can also run Gyro and board temperature):



Resetting the Nicla

Simple cases

Double-press the Reset button. That is usually enough.

Complex cases: flash the Edge Impulse firmware

Also the way to set the board up for machine learning with Edge Impulse.

1 Download the firmware

Download the pre-compiled Edge Impulse firmware .zip from the Edge Impulse documentation and extract it.

2 Enter bootloader mode

Double-press the Reset button. The **green LED flashes** when the board is ready.

3 Run the flash script

From the unzipped folder, run the script for your computer:

```
flash_windows.bat  
flash_mac.command  
flash_linux.sh
```



Accelerometer data + LED

Read the accelerometer data and print the data in the Serial Monitor

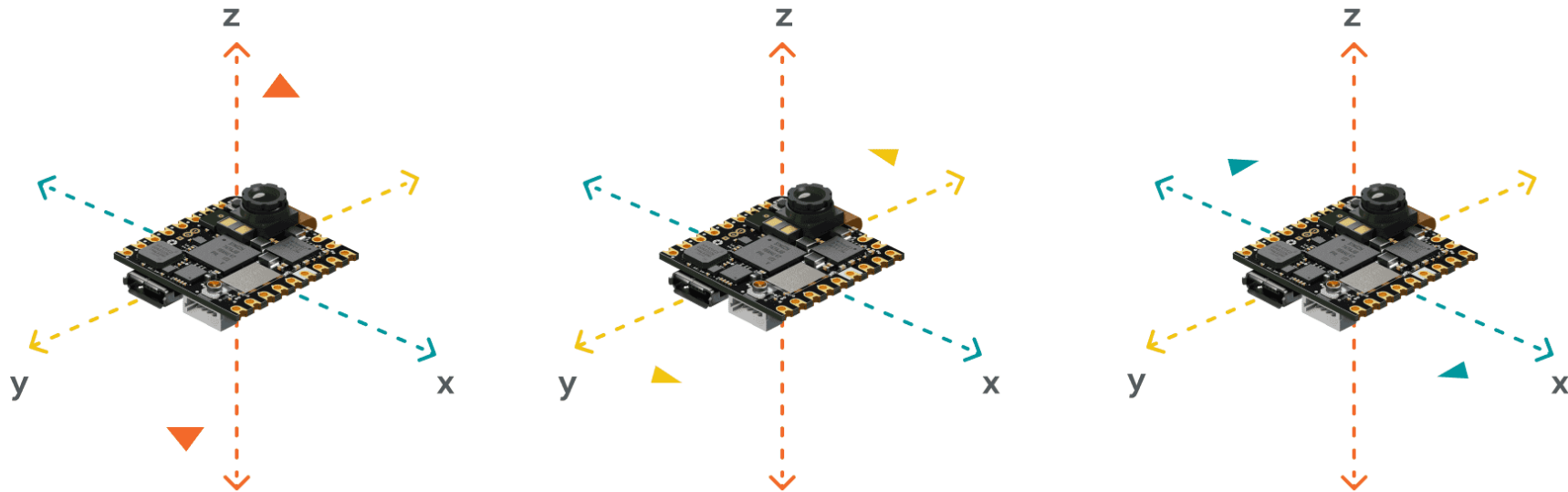
Exercise:

Blink the LED faster if a stronger acceleration is detected

Change the color of the LED according to the axis where the acceleration is stronger (RED for X, Green for Y and Yellow for Z)

Send a message using Serial Monitor when acceleration is stronger than a threshold value.

Read accelerometer data + LED



Proximity Detection+ LED

Read the Time of Flight (ToF) sensor data and print the data in the Serial Monitor

Exercise:

Blink the LED faster if the object is closer



Change the color of the LED according to the distance of the object (from Yellow to Red, for example)

Send a message using Serial Monitor when distance is shorter than a threshold value.



Eye Saver



Using the Time of Flight (ToF) sensor, the Eye Saver will check if a person is too close to the laptop screen. If so, a red LED will start blinking.

Thanks!

TinyML whispers low,
Code dances on micro souls,
AI blooms in small.